Receiving Sensus FlexNet Smart-Meter Transmissions with a Low-Cost SDR: Demodulation, Frame Structure, and Evidence of Payload Encryption

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***Abstract*—Sensus FlexNet is a licensed-band advanced metering infrastructure used by many North American electric utilities. Unlike the unlicensed ERT protocols that open-source tools already decode, FlexNet has no published receiver. We show that FlexNet's Normal uplink mode can be demodulated with a commodity RTL-SDR receiver. Read against a public 2009 Sensus specification, the mode the vendor calls "7-FSK" is frequency modulation by a $2{,}011\,\mathrm{Hz}$ subcarrier that carries one 16-QAM symbol per cycle. A seven-step receiver recovers tight 16-point constellations from every frame of a residential meter captured over $16\,\mathrm{h}$. Each frame holds 100 symbols: a two-symbol sync, eight blocks each led by a fixed per-meter header symbol, 88 payload symbols, and a two-symbol trailer. None of the three identifiers printed on the meter appears in the clear. The payload passes every randomness test applied, no convolutional code is visible under any symbol labelling, and the vendor documents AES-256 in CCM mode. We conclude that the payload is encrypted and stop there by design. The receiver is released as open-source software.**

***Index Terms*—advanced metering infrastructure, smart meters, software-defined radio, FlexNet, 16-QAM, protocol analysis**

I. Introduction

Utility smart meters report consumption over radio, and several common protocols are open to anyone with a receiver. Itron's ERT family, used by many gas and water meters, is decoded routinely with RTL-SDR dongles by tools such as `rtlamr`. Sensus FlexNet is different: it runs in licensed spectrum near $901\,\mathrm{MHz}$, its modulation is described only in vendor documents, and `rtlamr`'s maintainer lists it as unsupported and probably encrypted [1]. Home-automation users with FlexNet meters report no local way to read them [2].

We set out to answer three questions about one residential FlexNet meter, using only passive reception. How is the uplink modulated? What does a frame look like? Is the meter's data readable, or is it encrypted? This paper makes four contributions:

- an interpretation of the vendor's "7-FSK" specification as 16-QAM on an FM subcarrier, and a receiver built on it (Section V);
- a warning about an appealing but wrong 4-FSK reading of the same signal (Section V-C);
- the 100-symbol frame layout, separating protocol fields from per-meter fields (Section VI);
- a payload analysis that combines identifier searches, an exact search for the specified convolutional code, and randomness tests (Section VII).

TABLE I
FlexNet uplink modes in the 2009 specification

Mode	Description in the specification	kbit/s	Payload
Normal	7-FSK, 16-QAM, $\beta = 1.89$	8.045	28 B
C&I	as Normal, half baud	4.022	–
Priority	13-FSK, 256-MCM, $\beta = 2.62$	8.044	12 B
Double Density	13-FSK, full baud	16.089	–
mPass	GMSK	5	–

II. Scope and Ethics

All work was receive-only and concerned the author's own meter. Nothing was transmitted. FlexNet meters listen on a separate downlink, which was not touched. Once the evidence pointed to encryption, the work stopped. No key recovery or cryptanalysis was attempted, and this paper offers no help with either. A receiver also hears neighbouring meters; their frames were used only to tell protocol-fixed symbols from per-meter ones, and none of their contents are reported. The meter's printed identifiers and its on-air address are withheld. Rules on receiving encrypted transmissions vary by jurisdiction, and readers should check theirs.

III. Background

A 2009 Sensus document, "iCon Advanced Security FlexNet Integrated Display Transceiver, Modulation Specification," is public as an exhibit to an FCC experimental licence application [3]. Table I summarises the uplink modes it defines. It also specifies forward error correction by an interleaved rate-$\frac{1}{2}$ convolutional code with constraint length $K = 3$. FCC equipment grants for Sensus's electricity-meter transceivers list a 9K60F2D emission [4].

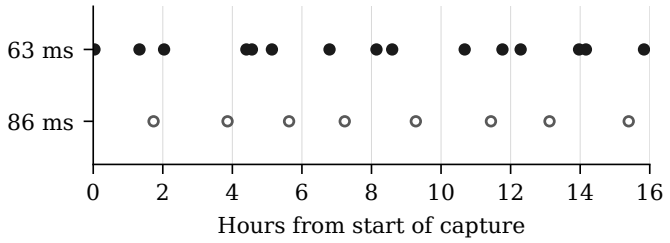


Fig. 1. Strong bursts over the first 16 h. The meter's 63 ms frames come roughly hourly on no fixed schedule; the 86 ms type is steadier.

IV. Capture

One RTL-SDR dongle on a Raspberry Pi 5 was tuned to 901.5 MHz at 2.4 MS/s, covering 900.3 MHz to 902.7 MHz, for 24 h. A burst detector tracks the noise floor in 1 ms blocks and treats any block more than 10 dB above it as part of a burst. For each burst it logs the time, duration, peak level and peak frequency, and saves the raw IQ with 20 ms of padding.

The band is busy. The first 16 h held 3,519 bursts, about 215 an hour, flat through the night. The meter under study is the strongest emitter on its channel, about 40 dB above the floor against 12 dB to 20 dB for other meters. It sent fifteen 63 ms frames near 901.06 MHz, 10 to 142 minutes apart (Fig. 1). A second frame type, 86 ms long near 901.01 MHz, arrived about every two hours at almost the same strength; we could not determine whether it comes from the same meter.

V. Demodulation

A. Reading the specification

The specification's Normal mode runs at 8.04455 kbit/s with a peak deviation of 3,808 Hz and a modulation index $\beta = 1.89$. These numbers fit one model exactly. The FM modulating waveform is a subcarrier at $f_s \approx 2{,}011$ Hz carrying one 16-QAM symbol $d_k \in \{\pm1, \pm3\} + j\{\pm1, \pm3\}$ per cycle:

$$f(t) = f_c + \kappa \,\mathrm{Re}\{d_k\, e^{j2\pi f_s t}\}, \quad \tfrac{k}{f_s} \le t < \tfrac{k+1}{f_s}. \qquad (1)$$

The symbol rate equals the subcarrier frequency, so the bit rate is $4f_s \approx 8{,}044$ bit/s and $\beta = 3808/2011 = 1.89$. The "7" in 7-FSK counts the instantaneous frequency levels such a waveform passes through, not its alphabet. Our reading is that 13-FSK carries two 16-QAM subcarriers ($16 \times 16 = 256$), but we observed only Normal mode.

B. Receiver

The receiver works on one saved burst at a time:

1) Mix the burst's peak frequency to 0 Hz and decimate to 240 kHz.
2) Low-pass at 40 kHz, FM-discriminate, and low-pass at 14 kHz.
3) Fit f_s on the bare-subcarrier preamble (about 12 ms) by maximising spectral energy over 2,000 Hz to 2,022 Hz in 0.02 Hz steps.
4) Form $z(t) = \big(\hat{f}(t) - \bar{f}\big)\, e^{-j2\pi f_s t}$ and rotate it so the preamble lies on the positive real axis.

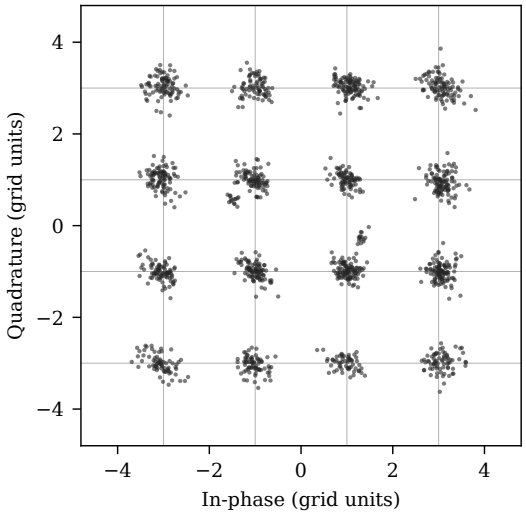


Fig. 2. Normalised symbol estimates from fourteen of the meter's frames, 1,400 points. Grey lines mark the grid levels ±1 and ±3.

5) Integrate and dump over one subcarrier period per symbol: $\hat{d}_k = \frac{1}{N}\sum_{n \in \text{period } k} z[n] \propto \kappa d_k/2$, sweeping the timing phase over 32 steps.
6) At each phase, fit a scale and a rotation to the 16-QAM grid and score the fit as the RMS distance to the nearest grid point in units of half the grid spacing. Keep the best phase.
7) Slice the in-phase and quadrature parts to $\{-3, -1, +1, +3\}$.

16-QAM looks the same every 90°, so symbol labels are defined only up to the rotation window the receiver searches. We centre that window on the residual rotation observed on real frames, about 64°, so every frame is labelled consistently. Fig. 2 shows the result for all fourteen demodulated frames from the meter. Their fit scores were 0.13 to 0.17, against about 0.44 at a wrong timing phase and 0.29 for random points.

C. A pitfall: the 4-FSK illusion

On a discriminator plot these bursts look like 4-level FSK at about 8,045 baud. Cyclostationary analysis finds a clock line there, and sampling at that rate yields plausible four-level symbols with an apparent preamble and sync word. That reading is wrong. No symbol-synchronous linear model fits: alphabets of 2 to 13 levels, rates from a quarter to twice 8,045 baud, and equalisers of up to nine taps all left a residual near 0.24 of a level spacing, where receiver noise alone would give 0.03. The clock line is exactly $4f_s$. Sampled four times per cycle, one 16-QAM symbol appears as the sequence $I, -Q, -I, Q$, which resembles four overlapping FSK symbols. A clean 2 kHz preamble tone with a clock line at four times its frequency is the sign to use a subcarrier receiver.

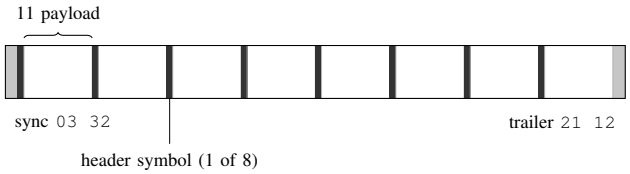


Fig. 3. The 100-symbol frame, to scale. Grey: identical on every meter observed. Black: identical in every frame of one meter, different between meters (32 bits). White: 88 payload symbols, different in every frame. Symbols are written as their in-phase and quadrature level indexes, 0–3 for -3 to $+3$.

VI. Frame Structure

All fourteen frames share one 100-symbol layout (Fig. 3). Aligning them and marking the positions that hold the same symbol in every frame leaves exactly twelve: two at each end and eight spaced twelve apart.

- **Sync and trailer.** Other meters end with the same trailer and share the second sync symbol; their first symbol was usually clipped by burst detection. These fields belong to the protocol.
- **Header.** Eight symbols, 32 bits, at positions $2{+}12k$. They were unchanged across $16\,\mathrm{h}$ and differ between meters, so they behave as a fixed address. They never count, so they are not a sequence number or a clock.
- **Payload.** The remaining 88 symbols, 352 coded bits, change almost completely between frames. Two frames from one neighbouring meter two minutes apart also agreed only at the header positions.

At 4 bits per symbol and a rate-$\frac{1}{2}$ code, 352 coded bits carry 176 information bits, 22 bytes. That is smaller than the specification's 28-byte payload, so either the code is not applied as specified or the frame carries less than the maximum. The $86\,\mathrm{ms}$ frames use the same modulation with a different layout: about 148 symbols, with a three-symbol header run every 18 symbols. We have not mapped them.

VII. Payload Analysis

A. Identifier search

The meter carries three printed numbers: the radio module's FlexNet ID, the utility's meter number, and the manufacturer's serial. We searched every frame for each one as binary-coded decimal and as plain binary at its natural width, 24 and 32 bits. Each encoding was also tried byte-swapped, least-significant bit first within bytes, and reversed. Every encoding was searched under 768 symbol-to-bit mappings. These are I and Q in either order, either axis negated, Gray or natural coding per axis, and all 24 bit orders within a symbol. No identifier appeared in the header. Across payloads one number matched once, in a single frame, against 0.25 matches expected by chance. A transmitted identifier would recur in every frame.

B. Search for the specified code

Finding a convolutional code from symbols normally means guessing the bit labelling, the interleaver and the generator polynomials together. We remove the labelling from the search. Let $f_j(s) \in \{0,1\}$ be bit j of the unknown label of symbol value $s \in \{0,\dots,15\}$, 64 unknowns in all. After de-interleaving into coded streams c_1 and c_2, every codeword of a rate-$\frac{1}{2}$ code with generators g_1 and g_2 satisfies the parity check

$$\bigoplus_k g_{2,k}\, c_1[m-k] \;\oplus\; \bigoplus_k g_{1,k}\, c_2[m-k] = 0, \qquad (2)$$

which is linear over GF(2) in the unknowns. For the correct configuration the true labelling is a non-trivial null vector; for a wrong one, hundreds of equations leave none. We count the nullspace dimension over the unknowns the equations mention and discount labellings that ignore the symbol, which satisfy any linear check.

The search covered every ordered pair of $K = 3$ generators drawn from $\{3,5,6,7\}$ (octal), block interleavers over the payload and spans up to 16 bits shorter, symbol-level and per-block interleavers, and up to two skipped symbols: 1,716 configurations on the 88-symbol payloads. A differential variant, which cancels any whitening sequence common to all frames, was also run. The same search on random symbols flagged 42 configurations as artefacts. No other configuration showed structure. On synthetic frames with a planted code, interleaver and arbitrary labelling, the search finds the planted configuration.

TABLE II
Payload statistics against random data of the same size

Statistic	Payload	Random, 95% range
χ^2 over 16 symbol values	15.27	5.69–27.53
Share of 1 bits	0.506	0.488–0.513
Runs test, z	0.55	-1.86–2.17
Max. |correlation|, lags 1–12	0.046	0.033–0.087
Longest run shared by two frames	5	4–5
Entropy per frame (bits/symbol)	3.875	3.850–3.896

C. Randomness

Table II compares the 1,232 payload symbols of the fourteen frames with 300 sets of uniformly random payloads of the same shape. Every statistic falls inside the random data's 95% range.

D. The vendor's description

A 2025 Sensus response to a utility's request for proposals [5] describes FlexNet encryption as AES-256 in CCM mode with CBC-MAC authentication, performed on the meter's radio module. It states that encryption is enabled per meter, is keyed to the meter's FlexNet ID, is optional for the utility, and "does not significantly increase the size of the packets." Newer meters use ECDH key agreement and ECDSA signatures; earlier ones use pre-shared keys.

VIII. Discussion

The measurements and the vendor's description agree. A clear per-meter address that lets the head end choose a key, followed by a structureless body with no size overhead, is

what AES-CCM looks like on the air. An unencrypted meter report sent hourly would show fixed fields, slowly changing counters, or repeated values between frames, and none appear. The absence of repeated blocks between frames also rules out ECB-style encryption.

Statistics alone cannot name a cipher: any modern cipher, or a strong per-frame whitening sequence, would produce the same results, so "AES" rests on the vendor's statement. Because encryption is optional per meter, other utilities' meters may transmit in the clear, and the tests described here would show it.

The study has limits. It covers one meter and the first $16\,\mathrm{h}$ of a $24\,\mathrm{h}$ capture. Frame length is as decoded; we cannot exclude an undecoded symbol at either edge, although the constant trailer suggests the frame end is found correctly. The $86\,\mathrm{ms}$ frames, the other modes and the downlink remain unexamined.

IX. Conclusion

FlexNet's Normal uplink is 16-QAM on a $2{,}011\,\mathrm{Hz}$ FM subcarrier and can be received cleanly with a commodity SDR. Its frames carry a fixed 32-bit per-meter address in the clear and an 88-symbol payload that behaves as ciphertext, consistent with the AES-256 CCM encryption the vendor documents. The practical route to a meter's own data remains the utility. The receiver, frame analysis and payload tests are released as `flexnet-rx` under the MIT licence; its test suite runs entirely on synthetic signals, so the methods can be checked without a meter.

Acknowledgment

The analysis and software were developed with the assistance of Claude, an AI model by Anthropic.

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